

COMP90007 Internet Technologies

Project 1: Network Analysis

Semester 1, 2026

Due date: **April 27th (Monday) 5:00pm**

1 Introduction

Weighting: 15% of the total marks for the subject. The weighting of each question is shown beside the question.

Tasks: The project focuses on evaluating network performance by measuring hop count, bandwidth, delay, and jitter. The key output of this project is a report answering the questions in Sections 2-4, which has to follow our provided format (refer to Section 5 for specific requirements). It is recommended that you perform these tasks in a consistent networking environment to minimise variance in your report. However, conducting them in different environments (e.g., some results from the university network and others from your home network) is also acceptable.

Along with this specification, we have also shared the following two documents:

- **Report Template:** This is the format you should use to answer the questions in our specification. Metrics captured can be shared across questions and sections but the corresponding explanations in each question should be **self-contained**, i.e. each question should be **answered separately** and needs to be clearly marked with the relevant question number, and **do not combine** answers for multiple questions.
- **Frequently Asked Questions (FAQ) document:** This will be the **primary resource** to refer to when you have any problems or have questions about this assessment. This document highlights key observations and common issues you may encounter while completing this assessment.

Important Notes:

- All work presented must be your original **individual** effort.
- All commands executed must be run **manually** on the terminal/command prompt. Automation scripts and/or processes to run these commands will not be accepted for this assessment and will result in significant mark deductions.
- As evidence of your work, when you run the commands in the following sections manually, you will need to take **screenshots of all the raw results** (every single execution of the commands) obtained and place them in the Appendix of your report. Including these screenshots is crucial for markers to verify results obtained by you. Page limits do not apply to the Appendix. Reports that fail to include all required screenshots will get significant mark deductions due to insufficient evidence of the captured results.
- All the plots and charts (e.g., scatter plots, bar charts, etc.) needed to answer questions should be placed in the main body of your report where you will explain your observations.
- Late submissions will attract a penalty of 10% per day (or part of a day).

2 Measuring the hop count (3 marks)

In this section, we will explore the number of intermediate hosts along a communication path to a remote server and analyse its relation to the physical geographical distance.

To count the number of hops taken to reach a destination host, the command `tracert` will be used (or its corresponding equivalent, depending on your operating system). This utility should be pre-installed on your operating system.

The utility can be invoked by launching a command line terminal and typing in the command. An **example output** of the `tracert` command (on OS X) and `tracert` command (on Windows) is as follows:

```
$ traceroute -nw 1 cis.unimelb.edu.au
traceroute to cis.unimelb.edu.au (128.250.37.164), 64 hops max, 52 byte packets
```

```

1  10.0.0.254  533.676 ms  1.063 ms  0.940 ms
2  58.96.2.205  27.872 ms  28.137 ms  28.293 ms
3  58.96.2.129  28.647 ms  28.577 ms  28.085 ms
4  218.100.78.33  28.299 ms  28.469 ms  28.332 ms
5  202.158.200.9  29.626 ms  28.871 ms  29.841 ms
6  202.158.210.26  31.320 ms  28.722 ms  29.135 ms
7  202.158.200.250  29.668 ms  29.096 ms  28.660 ms
8  * * *
9  * * *
10 * * *
11 128.250.37.130  957.521 ms  33.475 ms  29.891 ms
12 128.250.37.164  29.940 ms  29.260 ms  30.020 ms

```

In this section of the project, you are interested in the number of hops it takes to reach the destination server. In the example above, the number of hops to reach cis.unimelb.edu.au is 12. Your view may be slightly different based on your version of `tracert` but the information captured is similar.

Based on the number of measurements you will be taking, there are some useful command line parameters you can use to reduce the time it takes to gather results. The help documentation for the `tracert` utility can be accessed by running `man tracert` on OS X or `tracert /?` on Windows.

Specific task description

Please include all raw measurements in the Appendix.

- 2.1 Use Wireshark to capture the network trace generated when running the `tracert` command, only sending 1 packet instead of the default 3 packets, on any one of the hosts mentioned in Table 1. Explain `tracert`'s working using your chosen host, detailing the first 5 hops and the final hop of your packet capture. For the Wireshark capture, create a capture filter that catches only the `tracert` request and the response received without any other packets (like `dns`, `mdns`, etc.) and explain if this filter will capture other requests (non-`tracert` related) as well or not using examples (you do not need to run another trace for this).

Note: Windows' `tracert` does not allow sending 1 packet only per hop but MacOS/Linux/Unix does. To send 1 packet per hop in Windows, you will need to install [Windows Sub-System for Linux \(WSL\)](#).

- 2.2 Determine the hop count (using the default `tracert`/`tracert` 4 packets per hop configuration) for the following hosts given in Table 1. We ask that you find *one more public iperf server* other than the ones listed here to gather their results and use the *same 10 servers across all sections* of this specification to capture further metrics and answer questions.

Table 1: List of public `iperf` hosts

Host	Location
speedtest.awbb.net	USA
spd-desrv.hostkey.com	Germany
speedtest.wobcom.de	Austria
rychlost.poda.cz	Czech Republic
spd-fisrv.hostkey.com	Finland
speed.itgate.net	Italy
porto.speedtest.net.zon.pt	Porto
speedtest.masnet.ec	Ecuador
dal.speedtest.clouvider.net	USA

Determine the approximate geographical distance for the above hosts and plot a scatter chart of the hop count versus the approximate geographical distance from Melbourne. Do you observe a correlation or not? Please explain your rationale with respect to networking concepts.

The geographical distance can be calculated in a straight line from the request origin (Melbourne) to each respective host and hence will be an approximate distance only. Optionally, if you wish to find a more realistic measure of distance covered by their requests, you can calculate the distance travelled by looking at the origin of each hop noted by their `tracert` output.

Key Considerations:

- You do not need a VPN to run your tests against these hosts. If you do end up using a VPN, your geographical distance would be the distance from your device (the one you are running your tests on) to the geo-location of VPN servers and then to the end iperf hosts.
- You may use any scientific computing package or spreadsheet software to do your plotting, for example, Matlab, Microsoft Excel, etc.
- For finding out the physical geographical distance you may use any tool or application available online, for example, you may use a combination of: <https://db-ip.com> and <https://www.freemaptools.com/how-far-is-it-between.htm> or <https://www.site24x7.com/find-website-location.html> and <https://www.distancecalculator.net/> or anything of your choice. However, do make sure to **document it and provide the appropriate reference** to that application/ tool/ software used.

Note: The servers listed in Table 1 are public servers and are not maintained by the University of Melbourne, hence they are likely to go down at any point in time. Based on past experience, it would be advisable to **conduct your tests on these as soon as possible rather than leave it till the end** as there is a high probability that these servers might not be available at times and this cannot be used as an excuse for a late submission as we give a number of weeks for you to do the project. If these servers stop responding for a long time for some reason then please visit the link: <https://iperf.fr/iperf-servers.php> and find your own servers (anything that is responding) or feel free to find any public iperf server from the internet and note this in your report. Some alternate strategies worth exploring also include changing port numbers and trying to get the iperf metrics.

3 Measuring delay and jitter (4 marks)

In this section, you will be measuring the delay and jitter of the **hosts used in Section 2**, located in different geo-locations.

We will be using the `ping` utility, to measure the round-trip delay of packets. The `ping` utility should be pre-installed on all major operating systems. The standard deviation of the round-trip delay time will be taken as the value for *jitter* for this project.

The **standard deviation** measures the variation in a set of data. There are two ways of calculating standard deviation and different versions of `ping` can use either of the two formulas internally as part of the tool statistics. For this assessment, if we are calculating the standard deviation ourselves manually then we will use the following formula for our purpose:

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}$$

where $\bar{x}$ is the mean of the set of data. Details of this simple statistical measure can be found in many websites if you do not remember this from high school years. The FAQ document also provides an alternate formula to calculate standard deviation, depending on the type of data used. Please review it as well.

A sample output of the `ping` utility is shown below, but this output will vary depending on your operating system.

```
$ ping unimelb.edu.au
PING unimelb.edu.au (172.22.44.10): 56 data bytes
64 bytes from 172.22.44.10: icmp_seq=0 ttl=124 time=3.364 ms
64 bytes from 172.22.44.10: icmp_seq=1 ttl=124 time=3.416 ms
64 bytes from 172.22.44.10: icmp_seq=2 ttl=124 time=3.730 ms
^C
--- unimelb.edu.au ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 3.364/3.503/3.730/0.162 ms
```

In the output above, various statistics, including the mean and standard deviation, were calculated for you by the utility but given your version of the `ping` utility, you may or may not have exactly the same output. If you can capture approximate values with decimals as part of your `ping` output then it would be good to perform your own calculations to find the mean values and their standard deviation, to confirm the result.

Specific task description

For this section, you may want to consider the user-facing implications of high delay and high jitter in networking applications, especially for applications sensitive to the impact of high delay and high jitter. Please include all raw

measurements in the Appendix. For MacOS/ Linux users, please use ‘-c 4’ flag for ping to have 4 packets for each execution of ping. For example: ping www.google.com -c 4

- 3.1 Use Wireshark to capture the network trace generated when running the ping command on any one of the hosts mentioned in Table 1 (use 4 packets for a single ping execution). Explain each step of ping’s working to reach your chosen host by using the packets captured. For the Wireshark capture, create a capture filter that catches only the ping request and response without any other packets (like dns, mdns, etc) and explain if this filter will capture other requests (non-ping related) as well or not with examples.
- 3.2 Measure the round-trip delay for all the hosts used in Section 2 (including the hosts in Table 1, plus one more public iperf server selected by you). Collect **three** delay measurements (run this command 3 times - not 1 command gathering a few rows of ICMP responses) of each host and find the average round-trip delay and jitter by calculating the standard deviation manually or by using the standard deviation reported by your command output.

For all of the hosts, plot the scatter chart of average round-trip delay versus the approximate physical geographical distance to the server. Do the same with the jitter (i.e. jitter vs geographical distance). Outline the results in a table and ensure the calculations are included in the Appendix of the report.
- 3.3 From the two plots above, do you observe any correlation between delay and distance? Any correlation between jitter and distance? Why or why not? Explain your results comparatively with reference to the network environment in which you were collecting your results (this includes metrics like your download/upload speed, users sharing the network, load on network through other apps, etc.) and how does your networking environment influence your results obtained ? Examples required in your explanation.

4 Measuring the bandwidth-delay product (8 marks)

In this section, you will be measuring the bandwidth of different hosts in order to determine the bandwidth-delay product, using the results from the previous sections.

The utility that will be used to perform bandwidth measurements will be the iperf utility. This command line utility is available for download for all operating systems from <https://iperf.fr>. Alternatively you may choose to use the package manager for your relevant operating system.

There are two modes of operation in iperf. The server mode will host a server which will listen to incoming requests from a client. An iperf instance running in client mode will connect to the server, and packets will be exchanged and timed between the two hosts to calculate the bandwidth. In this project, we will be running iperf in *client mode*.

A sample output of iperf in client mode is shown below, noting the -c flag to designate operating in client mode.

```
$ iperf -c iperf.eenet.ee
-----
Client connecting to iperf.eenet.ee, TCP port 5001
TCP window size: 129 KByte (default)
-----
[ 5] local 10.0.0.5 port 51878 connected with 193.40.100.7 port 5001
[ ID] Interval      Transfer    Bandwidth
[ 5]  0.0-106.7 sec  128 KBytes  9.83 Kbits/sec
```

Note: Some servers in Table 1 may need to use the port 5002 or perhaps even port 80, so the command to override ports will be: “iperf3 -c speedtest.serverius.net -p 5002”. Also, **some iperf servers respond to iperf2, rest to iperf3** so please try to use both iperf2 and iperf3 to verify if the server is responsive to either version. If you have response from both iperf2 and iperf3 then you can pick iperf2 amongst them.

Specific task description

Please include all raw measurements in the Appendix as usual.

- 4.1 Define bandwidth-delay product. What does the bandwidth-delay product tell us about the data transmission capability of networks? Collect **three** sets of measurements (run this command 3 times) measuring the bandwidth of the public iperf hosts in Section 2 and find the mean bandwidth for each host.

- 4.2 Pick one host from Table 1 that is the **furthest away** from the location where you are running your tests and run the `iperf` command on it across four time slots - Morning (between 9-10am), Afternoon (1-2pm), Evening (6-7pm) and Night (11pm-12 midnight). Do you observe changes in your captured metrics? Explain why or why not by discussing the networking environment and usage patterns of the intermediate hosts/countries your request travels through at that time (using `traceroute`). Repeat the experiment and discussion for one more host from Table 1 which is **closest to** the location you are running your tests from.
- 4.3 Take the mean bandwidth and calculate the bandwidth-delay product in kilobits. You should use the mean round-trip delay time from your `ping` experiments as the delay time. Plot a bar chart for each host showing your results. You may wish to use a logarithmic scale, if appropriate.
- Explain your results making a comparative analysis with reference to your networking environment in which you performed your measurements. How do your results reflect upon your actual internet link speed and how does your network environment influence your results obtained? Examples required in your explanation.
- 4.4 List 2 major challenges you have encountered in your experiments with relevant evidence. For example, shared network, dynamic set of users, etc. Briefly describe these challenges and how you have addressed them *within* the constraints of the current experiments?

5 Project Administration

5.1 Getting help

If you have any questions, the discussion board (Ed Discussion) and FAQ document will be useful resources in resolving any issues. If your concern is a personal matter, then you should email the subject coordinator.

Any announcements made about the assignment in the lectures and any answers posted by the subject coordinator or the academic staff on the discussion board will be considered as part of the project specification. Additionally, please keep an eye on the Canvas announcements to any changes made to the project specification.

5.2 Report Format and Submission

The report will consist of all relevant data, graphs, answers and discussions from the experiments conducted in this project. The diagrams for answering questions, like plots, charts and so on must be placed in the main document, not the Appendix. All plots and figures must be appropriately labelled. Every diagram and/or raw measure used for a specific question must be referred to (using a consistent reference scheme, e.g. Fig. X) for markers to verify the result. The raw data must be included as **screenshots** in the Appendix of the report.

Any information obtained that is not of your own work must be cited. This report requires all citations and references to be in **IEEE** style.

The report must be submitted as a **PDF file** on Canvas via Project 1 page and no handwritten content will be accepted. Please include your name, student ID and login user name on the top of the first page. The report is to be formatted on A4 sized paper in 10 pt text, 1.5 line spacing, single column. The report should not exceed a **maximum of 12 pages** (excluding Appendix), else relevant penalties will apply. You are encouraged to follow the formatting scheme outlined in the **Report Template**.

All commands across sections would need to be run manually by typing the command yourself and getting screenshots of the command as well as the output received. You should not be automating this. Output captured outside of the terminal/command prompt (like .txt, .csv, etc) will not be accepted and lead to mark deductions.